



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

10-04-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-I)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	C	B	A	D	A	A	A	C
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	B	D	A	C	B	C	B	C	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	2	16	10	5	3	12	75	13	8	8

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	D	C	C	C	C	D	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	A	B	D	A	B	A	C	B	D
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	5	48	12	8	684	8	2	7	5	5

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	C	A	B	B	A	D	D	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	C	D	D	B	B	C	A	C	D	D
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	6	65	2	54	2	12	18	893	6	432

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (A)**

$$X = A.(A+B)$$

2. **Ans (A)**

Relative error for $x = \frac{d\ell}{\ell} \times \text{No. of repetition}$

$$x = \frac{0.3 \times 10^{-2}}{17.5} = 0.000171$$

Relative error for

$$y = \frac{0.1\text{cm}}{100\text{cm}} \times 17 + \frac{0.1}{51} = 0.0139$$

Relative error for

$$z = \frac{0.02\text{cm}}{30.48\text{cm}} \times 57 + \frac{0.02\text{cm}}{12.5\text{cm}} = 0.039$$

3. **Ans (C)**

$$w_1 = 300(2\pi), \quad w_2 = 900(2\pi)$$

$$\Delta w = w_2 - w_1 = \sqrt{3} \frac{R}{L}$$

$$\Rightarrow \text{Band width} = \frac{R}{L} = \frac{w_2 - w_1}{\sqrt{3}} = \frac{(600) 2\pi}{\sqrt{3}}$$

$$\text{Band width in Hz} = \frac{600}{\sqrt{3}} = 200\sqrt{3}$$

4. **Ans (B)**

$$\frac{mv^2}{r} = \frac{kze^2}{r^2} \quad \dots(1)$$

$$K. \varepsilon = \frac{1}{2}mv^2 = \frac{kze^2}{2r}$$

$$P. \varepsilon = \frac{-k(ze)e}{r}$$

$$\Rightarrow \text{Total energy } E = \frac{-kze^2}{2r} = \frac{-E_0}{n^2}$$

$$\Rightarrow r \propto n^2$$

$$\Rightarrow K. \varepsilon = \frac{kze^2}{2r}$$

$$v^2 \propto \frac{1}{n^2} \left(v \propto \frac{1}{n} \right) \quad w = \frac{v}{r} \propto \frac{1}{n \cdot n^2} \propto \frac{1}{n^3}$$

5. **Ans (A)**

$$V = iR \quad R = \rho \frac{\ell}{A},$$

$$\text{Volume : Constant} \Rightarrow \frac{\Delta \ell}{\ell} = -\frac{\Delta A}{A}$$

$$\frac{\Delta R}{R} = \rho \left(\frac{\Delta \ell}{\ell} - \frac{\Delta A}{A} \right)$$

$$\Rightarrow \frac{\Delta R}{R} = \rho \left(\frac{20\ell}{\ell} \right)$$

$$\Rightarrow \Delta R = \rho \left(\frac{2\ell}{\ell} \Delta \ell \right) \times R = (2\rho R) \rho$$

$$\Rightarrow V = i(R + \Delta R)$$

$$V = i(R + 2\rho R \text{ strain})$$

7. **Ans (A)**

$$\phi = (B_0 \sin \omega t) \pi r^2$$

$$\Rightarrow e = \left| \frac{d\phi}{dt} \right| = \pi r^2 B_0 \omega \cos \omega t$$

$$i = \frac{e}{R} = \frac{\pi r^2 B_0 \omega}{R} \cos \omega t$$

$$P = \int i^2 R dt$$

8. **Ans (A)**

For normal adjustment

$$m = \frac{v}{u} \left[1 + \frac{d}{fe} \right]$$

For objective

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_0} \Rightarrow \frac{v}{u} = 1 - \frac{v}{f_0}$$

$$\text{fo very small, } \Rightarrow \frac{v}{f_0} \gg 1$$

$$\Rightarrow \frac{v}{u} = -\frac{v}{f_0} = -\frac{\ell}{f_0}$$

$$\Rightarrow m = -\frac{\ell}{f_0} \left[1 + \frac{D}{fe} \right] = \frac{200}{10} \left[1 + \frac{250}{fe} \right]$$

$$\Rightarrow 400 = \frac{200}{10} \left[1 + \frac{200}{fe} \right]$$

9. **Ans (A)**

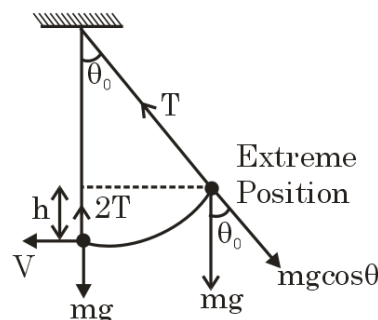
$$d\theta = \frac{B}{D} = \frac{\lambda}{d} = \frac{500 \times 10^{-9}}{4 \times 10^{-3}}$$

$$= \frac{5}{4} \times 10^{-4} \text{ rad} \times \frac{180^\circ}{\pi}$$

10. **Ans (C)**

40v, voltage across RL will be same as zener breakdown

11. **Ans (D)**



$$T - mg \cos \theta_0 = 0$$

$$T = mg \cos \theta_0$$

$$2T - mg = \frac{mv^2}{\ell}$$

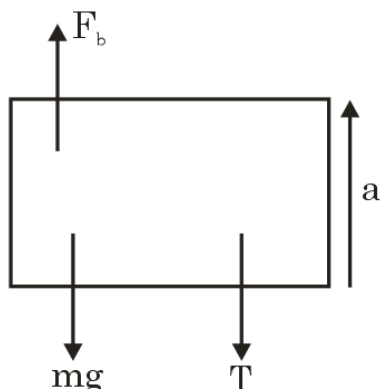
C.O.M.E.

$$\frac{1}{2}mv^2 = mgh \& h\ell(1 - \cos \theta_0)$$

$$2mg \cos \theta_0 - mg = 2mg(1 - \cos \theta_0)$$

$$\cos \theta_0 = \frac{3}{4}$$

12. Ans (B)



$$F_b = V \ell_2 \text{ gass}$$

$$= \frac{m}{P_b} \ell_L (g + a)$$

$$F_b - mg - T = ma$$

$$T = F_{ub} - m(g + a)$$

$$= m(g + a) \frac{\ell_L}{\ell_b} - m(g + a)$$

$$T = m(g + a) \left(\frac{\ell_L}{\ell_b} - 1 \right)$$

$$= 1(10 + 1) \left(\frac{1}{0.8} - 1 \right)$$

$$= \frac{11}{4} = 2.75 \text{ N}$$

13. Ans (D)

$$\frac{F}{A} \alpha \frac{du}{dy} \quad u = \alpha \left[\frac{y}{n} - 2 \left(\frac{y}{h} \right)^2 \right]$$

$$\frac{du}{dy} = \alpha \left[\frac{1}{h} - 4 \frac{y}{h^2} \right]$$

Stren at fixed plete $y = 0$

$$\frac{F}{A} = x \alpha \left[\frac{1}{h} - \frac{4 \times 0}{h^2} \right] = \frac{x \alpha}{h}$$

14. Ans (A)

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2} = \frac{4 - 1}{4 + 1} g = 6 \text{ m/sec}^2$$

$$\text{S.K.E.} = \left(\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \right) - \left(\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \right)$$

$$= m_1 g h_1 - m_2 g h_2 = (m_1 - m_2) g h$$

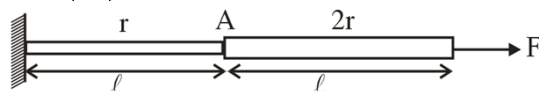
h is displacement in 3rd second

$$h = u + \frac{a}{2} (2n - 1)$$

$$= 0 + \frac{1}{2} \times 6 \times (2 \times 3 - 1) = 15 \text{ m}$$

$$\text{S.K.E.} = (4 - 1) \times 10 \times 15 = 450 \text{ J}$$

15. Ans (C)



$$\frac{F}{A_1} = \lambda \frac{\Delta \lambda_1}{\ell}, \quad \frac{F}{A_2} = \lambda \frac{\Delta \lambda_2}{\ell}$$

$$\Delta \lambda_1 + \Delta \lambda_2 = 10 \text{ mm}$$

$$\frac{F \ell}{A_1 \lambda} + \frac{F \ell_2}{A_2 \lambda} = 10 \text{ mm}; \quad \frac{F \ell}{A_1 \lambda} + \frac{F \ell_2}{4 A_1 \lambda} = 10 \text{ mm}$$

$$\frac{F \ell}{A_1 \lambda} = 8 \text{ mm}; \quad \Delta \ell = 8 \text{ mm}$$

OR

$$K_1 = \frac{A_1 \lambda}{\ell}, \quad K_2 = \frac{A_2 \lambda}{\ell} = 4 K_1$$

$$K_{er} = \frac{k_1 k_2}{k_1 + k_2} = \frac{k_1 \times 4 k_1}{k_1 + 4 k_1} = \frac{4}{5} k_1$$

$$F = k_1 \Delta \lambda_1 = K_{er} \Delta \lambda$$

$$k_1 \Delta \lambda_1 = \frac{4}{5} k_1 \times 10 \text{ m} \quad \Delta \lambda_1 = \frac{4}{5} \times 10 = 8 \text{ mm}$$

16. Ans (B)

$$(50 + 50 + 95 + 5) \times 2 = 1950 + 5(V + u)$$

$$400 = 200V + 20$$

$$V = \frac{300}{200} = 1.8 \text{ m/sec}$$

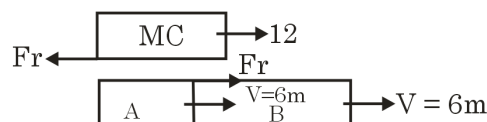
$$t = \frac{S_{rel}}{U_{rel}} = \frac{4m}{4m/5u} = 1 \text{ sec}$$

$$\text{Displacement} = 1.8 \times 1 = 1.9 \text{ m}$$

$$= 190 \text{ cm}$$

17. Ans (C)

Collision between A & B
 $m \times V = 2 m V'$
 $V' = V/2 = 6 \text{ m/sec}$,
 Velocity of A & D just after collision A & D
 & Velocity of block C just effect collision is 12 m/sec



$$a_c = \frac{F_r}{m} = \frac{-Hmg}{m} = -Hg \leftarrow$$

$$a_{A+B} = \frac{F_r}{2m} = \frac{Hmg}{2m} = \frac{Hg}{2} \rightarrow$$

$$u_{rel} = a_c + a_b = Hg - \frac{Hg}{2} = -\frac{3Hg}{2} = -3 \text{ m/sec}^2$$

Finally $u_{rel} = 0$

$$v_{rel}^2 = u_{rel}^2 + 2 a_{rel} s_{rel}$$

$$0^2 = (12 - 6)^2 + 2 \times -3 \times \ell$$

$$\ell = 6 \text{ m}$$

18. Ans (B)

$$\frac{w_0}{16} = \tan 37 = \frac{3}{4}$$

$$w_0 = 12 \text{ rad/sec}$$

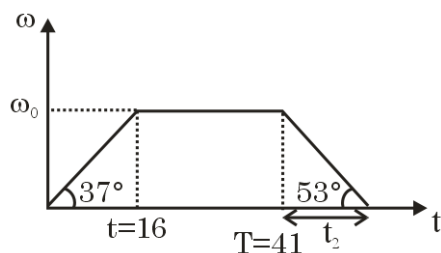
$$\frac{w_0}{t_2} = \tan 53 = \frac{4}{3}$$

$$\frac{12}{t_2} = \frac{4}{3} \rightarrow t_2 = 9 \text{ sec}$$

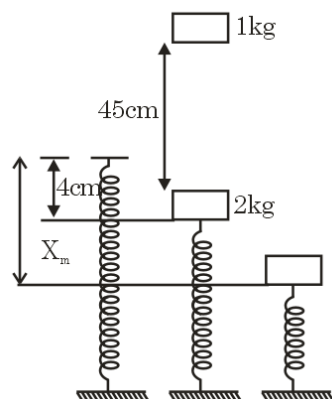
Angle turn by particle is total time =

$$\text{Area under curve} = \frac{1}{2} (50 + 25) \times 12 = 450$$

$$\langle w \rangle = \frac{\theta}{t} = \frac{450}{50} = 9 \text{ rad/sec}$$



19. Ans (C)



$$2g = k x_0 = k \frac{4}{100}$$

$$k = 500 \text{ N/m}$$

$$V = \sqrt{2gh} = \sqrt{2 \times 10 \times 0.45} = 3 \text{ m/sec}$$

$$1 \times 3 = (2+1)V$$

$$V = 3 \text{ m/sec}$$

at maximum compression

$$3y \times 2y + \frac{1}{2} \times 500 [(0.04)^2 - (0.04 + y)^2]$$

$$= -\frac{1}{2} \times 3 \times 1^2$$

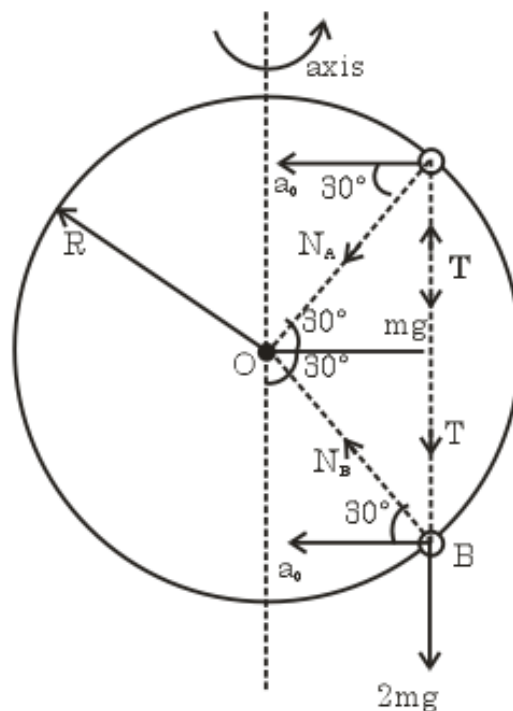
$$250y^2 - 10y - \frac{3}{2} = 0$$

$$y = \frac{10 \pm \sqrt{100 + 4 \times 250 \times \frac{3}{2}}}{500}$$

$$y = \frac{10 \pm \sqrt{1600}}{500} = \frac{10 \pm 40}{500} = 0.1$$

$$\text{Maximum Compression} = 10 + 4 = 14 \text{ cm}$$

20. Ans (B)



For Bead A $\Rightarrow a_A$ towards axis which can be provided by

N_A only as T & $m_n g$ is vertical

$$N_A \cos 30 = mw^2 r = mw^2 R \cos 30^\circ$$

$$N_A = mw^2 R$$

$$T = mg + N_A \sin 30^\circ$$

$$T = mg + \frac{mw^2 R}{2}$$

For Bead 'B' a_B is towards axis which can only be provided by N_B

$$N_B \cos 30^\circ = 2mw^2 r$$

$$N_B \cos 30 = 2mw^2 R \cos 30^\circ$$

$$N_B = 2mw^2 R$$

$$T + 2mg = N_B \sin 30^\circ$$

$$mg + \frac{mw^2 R}{2} + 2mg = 2mw^2 R \times \frac{1}{2}$$

$$3mg = \frac{mw^2 R}{2}$$

$$w = \sqrt{\frac{6g}{R}} = \sqrt{\frac{6 \times 10 \times 3}{5}} = 6 \text{ rad/sec}$$

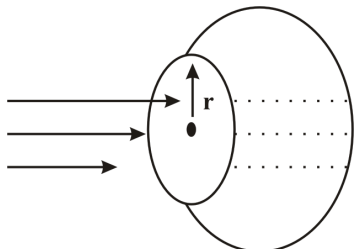
PART-1 : PHYSICS

SECTION-II

1. **Ans (2)**

$$(\pi r^2) (2\rho V^2) = 2(2\pi r) T$$

$$\Rightarrow \vartheta = \sqrt{\frac{2T}{\rho r}} \Rightarrow X = 2$$



2. **Ans (16)**

$$\phi_1 = 0, \phi_2 = EA = 4(2^2) = 16$$

$$x = 2, E = 4\hat{i}$$

3. **Ans (10)**

$$B = \mu_r \mu_0 n i =$$

$$= 10000 \times 4\pi \times 10^{-7} \times 500 \times 5 = 10\pi$$

$$X = 10$$

4. **Ans (5)**

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$= 1.1 \times 10^7 \times (36)^2 \left\{ \frac{1}{4} - \frac{1}{9} \right\}$$

$$\lambda = 5.05 \text{ \AA}$$

5. **Ans (3)**



$$V = 3t^2$$

$$\frac{L di}{dt} = 3t^2 \quad u = \frac{1}{2} Li^2$$

$$\int L di = \int 3t^2 dt \quad \frac{du}{dt} = \frac{Li di}{dt}$$

$$Li = \frac{3t^3}{3} + C \Rightarrow \frac{du}{dt} = (3t^2) \left(\frac{t^3}{L} \right)$$

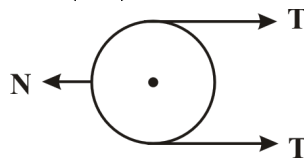
$$t=0, i=0 \Rightarrow C=0$$

$$\Rightarrow i = \frac{t^3}{L} \quad \frac{du}{dt} = \frac{3}{L} t^5$$

$$t=1 \quad \frac{du}{dt} = \frac{3}{2} = 1.5 \text{ J/s}$$

$$\Rightarrow x = 1.5$$

6. **Ans (12)**



$$2T - N = m\omega^2 R_{cm}$$

$$= \mu \times \pi R \times \left(\frac{V}{R} \right)^2 \times \frac{2R}{\pi}$$

$$2T - N = 2\mu v^2$$

$$N = 2T - 2\mu v^2$$

$$= 2 \times 10 - 2 \times 1 \times 4 = 12 \text{ N}$$

7. **Ans (75)**

$$\frac{P_2}{P_1} = \frac{V_2}{V_1} \Rightarrow P_2 V_1 = P_1 V_2$$

$$W = P_2 (V_2 - V_1) - P_1 (V_2 - V_1)$$

$$= nRT_2 + nRT_1 - P_2 V_1 - P_1 V_2$$

$$P_2 = \frac{nRT_2}{V_2}, P_1 = \frac{nRT_1}{V_1}$$

$$\frac{P_2}{P_1} = \frac{T_2}{T_1} \times \frac{V_1}{V_2} = \frac{V_1}{V_2}$$

$$\Rightarrow \frac{V_1}{V_2} = \sqrt{\frac{T_2}{T_1}}$$

$$\Rightarrow P_2 V_1 = P_1 V_2 = nR \sqrt{T_1 T_2}$$

$$\Rightarrow W = nR (\sqrt{T_1} - \sqrt{T_2})^2$$

$$= 1 \times \frac{25}{3} (20 - 17)^2 = 75 \text{ J}$$

8. **Ans (13)**

$$x = 3x_1 + 4x_2$$

$$a = 3a_1 + 4a_2 \quad \dots(1)$$

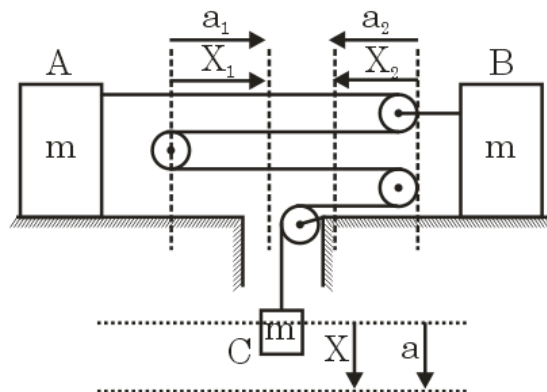
$$3T = ma_1 \quad \dots(2)$$

$$4T = ma_2 \quad \dots(3)$$

$$mg - T = ma \quad \dots(4)$$

$$\text{from (1), (2), (3) and (4)}$$

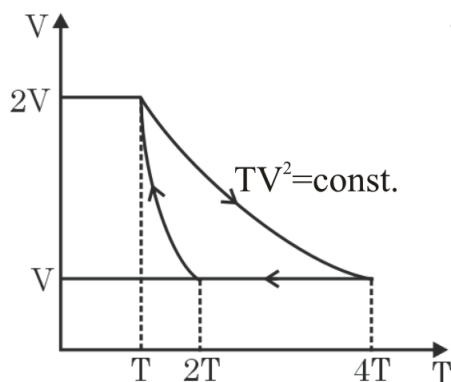
$$a = 2g/13$$



9. Ans (8)

$$P_{\max} = \frac{nRT_{\max}}{V_{\min}} = \frac{nR \times 4T}{V}$$

$$P_{\min} = \frac{nRT_{\min}}{V_{\max}} = \frac{nRT}{2V} \quad \frac{P_{\max}}{P_{\min}} = \frac{8}{1}$$



$$T_{\max} = T_A = 4T, T_{\min} = T_C = T$$

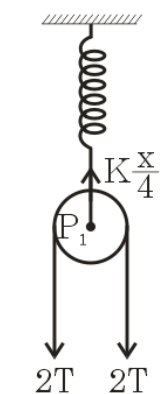
$$V_{\max} = V_C = 2V, V_{\min} = V_B = V_A = V$$

10. Ans (8)

From equilibrium position if block displaced downward by x pulley P_2 and P_1 moves $\frac{x}{2}$ and $\frac{x}{4}$ downward and spring further stretched by $\frac{x}{4}$

For P_1

$$4T = k \frac{x}{4}; T = \frac{K}{16}x$$



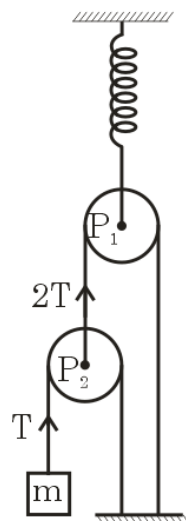
For block



$$F_{\text{net}} = T = -\frac{K}{16}x$$

$$\text{Time period} = 2\pi \sqrt{\frac{m}{K/16}}$$

$$= 8\pi \sqrt{\frac{m}{K}}$$



PART-2 : CHEMISTRY

SECTION-I

1. Ans (B)

$$r \propto n^2 \Rightarrow \frac{r_2}{r_1} = \frac{n_2^2}{n_1^2}$$

$$\Rightarrow \frac{x}{r_1(L_i^{+2})} = \frac{3^2}{1^2} \Rightarrow r_1(L_i^{+2}) = \frac{\pi}{9}$$

$$\Rightarrow \because 2\pi r = n\lambda \Rightarrow \because 2\pi \times \frac{x}{9} = 1 \times \lambda$$

$$\lambda = \frac{2\pi x}{9}$$

2. Ans (A)



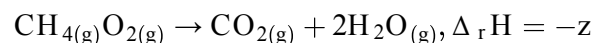
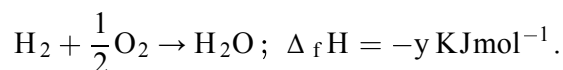
wt. of sample = 6.3g, Vol. = 250 mL.

$\Rightarrow$ Equivalents of oxalic acid eq. of NaOH

$$2 \times \frac{6.3}{126} \times \frac{1000}{250} \times 10 = 0.1 \text{ N}$$

$$V = 40 \text{ mL.}$$

3. Ans (D)



$$-z = (x + 2 \times -y) - \Delta_f H_C(\text{H}_4)$$

$$\Delta_f H_C(\text{CH}_4) = (-x - 2y + z)$$

4. Ans (C)

$$[\text{HQ}] = 0.1 \text{ M}; \text{pH} = 3 \Rightarrow [\text{H}^+] = 10^{-3} \text{ M}$$

$$K_a = \frac{10^{-3} \times 10^{-3}}{(0.1 - 10^{-3})} = 10^{-5}$$

5. **Ans (C)**

Hydrogen can get reduced so it can be have as oxidising agent.

6. **Ans (C)**

Factual (NCERT Based)

7. **Ans (C)**

Higher the solvation of ions, slower will be their movement

∴ Conductivity depends on solvation

8. **Ans (D)**

Adsorption of ions from solution is responsible and not the absorption.

9. **Ans (A)**

V_2O_5 and Cr_2O_3 are amphoteric oxides.

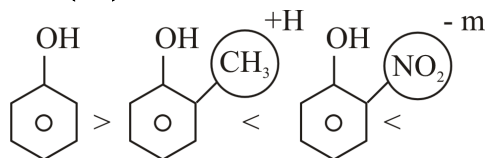
V_2O_4 and CrO are Basic oxides.

Mn_2O_7 and CrO_3 are Acidic oxides.

10. **Ans (A)**

During besbezerisation slag is formed and self reduction takes place.

11. **Ans (A)**

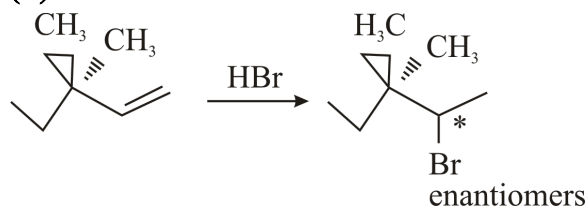


12. **Ans (A)**

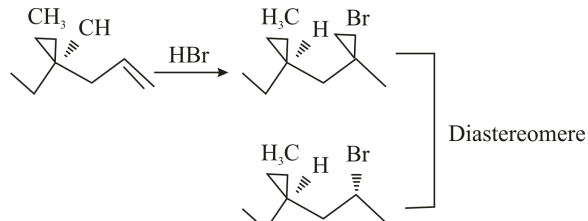
Check with R/S configuration

13. **Ans (B)**

(A)



(B)



(C) Antimerkoni Koff's addition, only one product

(D) Antimerkoni Koff's addition, only one product

14. **Ans (D)**

A, B and C will give Hoffmann alkene

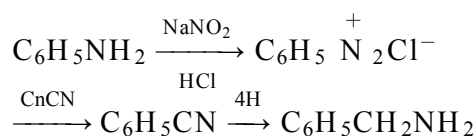
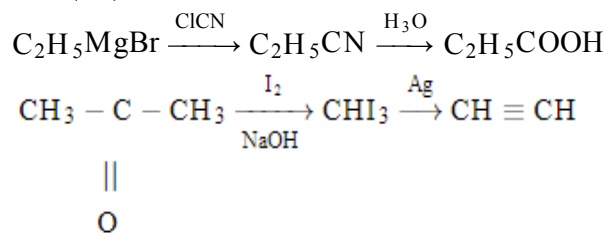
15. **Ans (A)**

HCHO will get oxidized due to more reactivity for nucleophilic attack

16. **Ans (B)**

OH will decide the potion pf upcoming electrophile

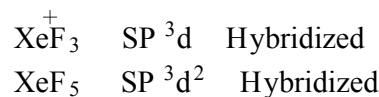
17. **Ans (A)**



18. **Ans (C)**

L-Glucose is enantiomers of D-Glucose

19. **Ans (B)**



20. **Ans (D)**

$$\Delta H_1 \rightarrow -ve$$

$$\Delta H_2 \rightarrow -ve$$

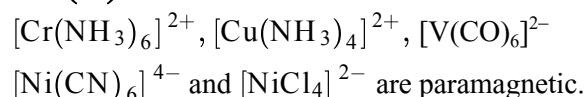
$$\Delta H_3 \rightarrow -ve$$

$$\Delta H_4 \rightarrow +ve$$

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (5)**



2. **Ans (48)**

$$K = \frac{2.303}{32} \log \left(\frac{100}{100-1} \right) = \frac{2.303}{t_{99.9}} \log \left(\frac{100}{100-99.9} \right)$$

$$\frac{2.303}{32} \times 2 = \frac{2.303}{t_{99.9}} \times 3$$

$$\Rightarrow 48 \text{ minutes}$$

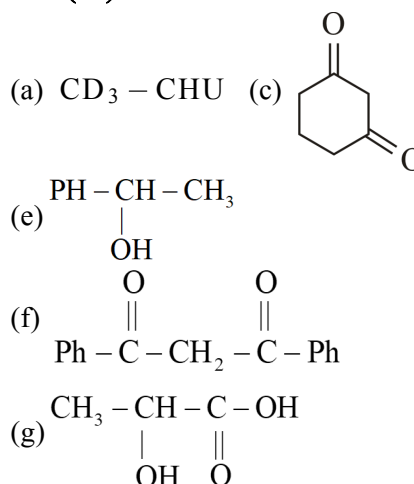
3. **Ans (12)**
160 ppm of Ca^{+2} ions in solution.
 $\Rightarrow 160 \text{ g of } \text{Ca}^{+2} \text{ in } 10^6 \text{ g solution.}$
 $\Rightarrow \therefore \text{ moles of } \text{Ca}^{+2} \text{ ions} = \frac{200}{40} = 4 \text{ mol.}$
 $\Rightarrow \therefore \text{ Moles of } \text{H}^+ = 2 \times 5 = 10 \text{ mol}$
(to maintain charge neutrality double moles of H^+)
are required.
 $\Rightarrow \text{ Density of solution} = 1 \text{ g mL}^{-1} (\because \text{ it is very dilute})$
 $\text{Vol. of solution} = 10^6 \text{ mL}$
 $[\text{H}^+] = \frac{10}{10^6} \times 1000 = 10^{-2} \text{ M}$
 $\text{pH} = 2$
 $\therefore \text{ pOH} = 12$

4. **Ans (8)**
No of P-O-P Bonds in $\text{P}_4\text{O}_6 = 6$
No of S-O-S Bonds in λ from of $\text{SO}_3 = 3$
No of B-O-B Bonds in B or ax = 5
Ans. 11 - 3 = 8

5. **Ans (684)**
 $\Rightarrow \therefore \text{ At normal conditions difference below}$
boiling and freezing point is 100°C .
 $\Rightarrow \therefore \Delta T_b + \Delta T_f = 4.76^\circ\text{C}$
 $\Delta T_b = K_b \times \text{Molality}$
and $\Delta T_f = K_f \times \text{Molality}$
 $\therefore \Delta T_b + \Delta T_f = K_b \times \text{Molality} + K_f \times \text{Molality}$
 $\Delta T_b + \Delta T_f = (K_b + K_f) \times \text{Molality}$
 $4.76 = (0.52 + 1.86) \times \frac{\omega}{342} \times \frac{1000}{1000}$
 $4.76 = 2.38 \times \frac{\omega}{342}$
 $\omega = 684 \text{ gm}$

6. **Ans (8)**
(i) IF_5 (SP^3)
(ii) ICl_4^- (sp^3d^2)
(iii) XeO_2F_2 (SP^3d)
(iv) NH_2^- (sp^3)
(v) BCl_3 (SP^2)
(vi) BeCl_2 (SP)
(vii) AsCl_4^+ (SP^3)
(viii) BH_4^- (sp^3)
(ix) NO_2^- (SP^2)
(x) ClO_2^+ (SP^2)

7. **Ans (2)**
Buna - N and Buna - S
8. **Ans (7)**
(i), (ii), (iii), (iv), (v), (viii), (ix)
9. **Ans (5)**
b, d, e, f, g
10. **Ans (5)**



PART-3 : MATHEMATICS

SECTION-I

1. **Ans (A)**
$$\frac{(\tan \frac{3\pi}{8} - \tan \frac{\pi}{8})}{\tan(\frac{3\pi}{8} - \frac{\pi}{8})} + \frac{(\tan \frac{5\pi}{8} - \tan \frac{3\pi}{8})}{\tan(\frac{5\pi}{8} - \frac{3\pi}{8})}$$

$$+ \frac{(\tan \frac{7\pi}{8} - \tan \frac{5\pi}{8})}{\tan(\frac{7\pi}{8} - \frac{5\pi}{8})} + \frac{(\tan \frac{9\pi}{8} - \tan \frac{7\pi}{8})}{\tan(\frac{9\pi}{8} - \frac{7\pi}{8})}$$

$$= \tan \frac{9\pi}{8} - \tan \frac{\pi}{8} = 0$$
2. **Ans (B)**
Since line and plane are parallel
 $\therefore \text{ Equation of plane : } x + 5y + z = 5$
 $\therefore d = \frac{|2 - 10 + 3 - 5|}{\sqrt{27}}$
$$= \frac{10}{3\sqrt{3}}$$
3. **Ans (C)**
 $\sin x + 2 \cos x \cos \theta = 2$
$$= -\sqrt{1 + 4\cos^2 \theta} \leq 2 \leq \sqrt{1 + 4\cos^2 \theta}$$

$$\sin^2 \theta \leq \frac{1}{4}$$

4. **Ans (A)**

As denominator is not converging

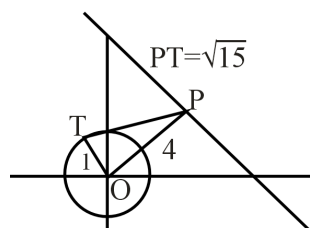
The limit is of the form $\frac{\text{finite}}{\text{infinite}}$

$\therefore$ Value of limit = 0

5. **Ans (B)**

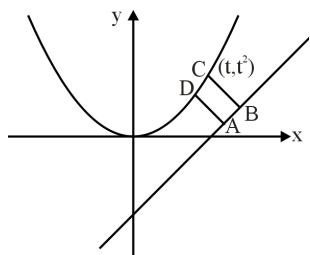
$$[\bar{a} \bar{b} \bar{a} \times \bar{b}] = |\bar{a} \times \bar{b}|^2 = 3$$

6. **Ans (B)**



$$PT^2 = OP^2 - r^2 \\ = 16 - 1 = 15$$

7. **Ans (A)**



$$BC = \frac{|2t - t^2 - 17|}{\sqrt{5}} = a$$

$$\Rightarrow t^2 - 2t + 17 - \sqrt{5}a = 0 \quad \dots(i)$$

if $C(t_1, t_1^2)$, $D(t_2, t_2^2)$ then

t_1, t_2 are root of equation (i) and $CD = a$

$$\Rightarrow t_1 + t_2 = 2, t_1 t_2 = 17 - \sqrt{5}a$$

$$\Rightarrow a^2 = (t_1 - t_2)^2 + (t_1^2 - t_2^2)^2$$

$$a^2 = 4 - 4(17 - \sqrt{5}a) + 4(4 - 4(17 - \sqrt{5}a))$$

$$a^2 = 20 - 20(17) + 20\sqrt{5}a$$

$$\Rightarrow a^2 - 20\sqrt{5}a + 320 = 0$$

$$\Rightarrow a = 16\sqrt{5} \text{ or } a = 4\sqrt{5}$$

minimum area is when

$$a = 4\sqrt{5} \Rightarrow a^2 = 80$$

8. **Ans (D)**

By using condition of tangency, we get $4h^2 = 3k^2 + 2$

$\therefore$ Locus of $P(h, k)$ is $4x^2 - 3y^2 = 2$ (which is hyperbola.)

$$\text{Hence } e^2 = 1 + \frac{b^2}{a^2} \Rightarrow e = \sqrt{\frac{7}{3}} \text{ Ans.}$$

9. **Ans (D)**

$$\text{We have } 4y^2 + (4x)y + (x+6) = 0$$

As $y \in \mathbb{R}$, so discriminant ≥ 0 .

$$\Rightarrow (4x)^2 - 4 \times 4(x+6) \geq 0$$

$$\Rightarrow x^2 - (x+6) \geq 0 \Rightarrow x^2 - x - 6 \geq 0$$

$$\Rightarrow (x-3)(x+2) \geq 0$$

$$\text{Hence } x \in (-\infty, -2] \cup [3, \infty)$$

10. **Ans (B)**

$$\text{Given } (\cot^{-1}x)(\tan^{-1}x) + \left(2 - \frac{\pi}{2}\right) \cot^{-1}x - 3$$

$$\tan^{-1}x - 3 \left(2 - \frac{\pi}{2}\right) > 0$$

$$(\cot^{-1}x - 3) \left(\tan^{-1}x + 2 - \frac{\pi}{2}\right) > 0$$

$$\left(\text{As } \tan^{-1}x - \frac{\pi}{2} = -\cot^{-1}x\right)$$

$$\Rightarrow (\cot^{-1}x - 3)(2 - \cot^{-1}x) > 0$$

$$\Rightarrow (\cot^{-1}x - 3)(\cot^{-1}x - 2) < 0$$

$$\Rightarrow 2 < \cot^{-1}x < 3$$

$$\Rightarrow \cot 3 < x < \cot 2 \quad (\text{As } \cot^{-1}x \text{ is a decreasing function.})$$

$$\text{Hence } x \in (\cot 3, \cot 2)$$

11. **Ans (C)**

Consider the expansion of $(1+x)^{99}$ and put $x = i$ and equate the real part to get option C

12. **Ans (D)**

$$f(x) = x^4 - 2x^3 + 3x^2 - 4x + 1 = 0$$

$$f'(x) = 4x^3 - 6x^2 + 6x - 4$$

$$= 2(x-1)(2x^2 - x + 2)$$

$$\Rightarrow f'(x) = 0 \text{ only at } x = 1$$

$$\text{also } f''(x) = 12x^2 - 12x + 6$$

$$= 6(2x^2 - 2x + 1) > 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow f \text{ is concave up}$$

$$f''(1) > 0 \Rightarrow x = 1 \text{ is minima}$$

$$\text{also } f(0) = 1, f(1) = -1$$

$$\Rightarrow \text{two positive and two non real complex roots}$$

13. Ans (D)

$$T_k = \frac{(k+2)\sqrt{k} - k\sqrt{k+2}}{k(k+2)^2 - k^2(k+2)}$$

$$= \frac{(k+2)\sqrt{k} - k\sqrt{k+2}}{2k(k+2)}$$

$$= \frac{1}{2} \left[\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+2}} \right]$$

$$T_1 = \frac{1}{2} \left[\frac{1}{\sqrt{1}} - \frac{1}{\sqrt{3}} \right]$$

$$T_2 = \frac{1}{2} \left[\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{4}} \right]$$

$$T_3 = \frac{1}{2} \left[\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{5}} \right] \text{ and so on}$$

$$\therefore \text{as } k \rightarrow \infty \text{ sum}$$

$$= \frac{1}{2} \left[1 + \frac{1}{\sqrt{2}} \right] = \frac{1+\sqrt{2}}{2\sqrt{2}} = \frac{\sqrt{1}+\sqrt{2}}{\sqrt{8}}$$

$$\Rightarrow a+b+c = 11$$

14. Ans (B)

$$\therefore f'''(x) = f'''(x-2)$$

$$\Rightarrow f \text{ is a cubic polynomial function}$$

$$\therefore \text{Let } f(x) = ax^3 + px^2 + qx + r$$

$$\text{now use } f(x) - f(x-2) = 4x^2 - 4x + 1$$

$$\text{compare the coefficients to get}$$

$$a = 2/3, p = 1, q = -1/6,$$

$$\text{hence } p + q = 5/6$$

15. Ans (B)

$$\text{Required probability} = \frac{\text{length of favourable interval}}{\text{length of total interval}}$$

$$P = \frac{2}{3}$$

$$\text{Hence } a = 2, b = 3$$

16. Ans (C)

If $(a, b) \in R$
then $(b, a) \in R$
Hence it is a symmetric relation

17. Ans (A)

Given: $N = 15$
 $\Sigma x^2 = 2830, \Sigma x = 170$
 One observation 20 was replaced by 30, then
 $\Sigma x^2 = 2830 - 400 + 900 = 3330$
 $\Sigma x = 170 - 20 + 30 = 180$
 $\therefore \text{Variance, } \sigma^2 = \frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N} \right)^2$
 $= \frac{3330}{15} - \left(\frac{180}{15} \right)^2$
 $= 222 - 144 = 78$

18. Ans (C)

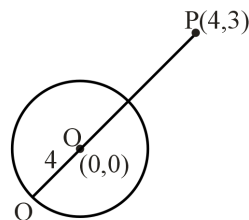
$$s = \sim(p \wedge q) \vee \sim(q \Leftrightarrow p)$$

p	q	$\sim(p \wedge q)$	$\sim(p \Leftrightarrow q)$	s
T	T	F	F	F
T	F	T	T	T
F	T	T	T	T
F	F	T	F	T

19. Ans (D)

Consider a function $g(x) = x f(x)$
 Obviously g is continuous in $[0, 1]$ and
 differentiable in $(0, 1)$
 As $f(1) = 0$
 $\therefore g(0) = 0 = g(1)$
 Hence Rolle's theorem is applicable for g
 $\therefore \exists$ some $c \in (0, 1)$ such that
 $g'(c) = 0$
 $c f'(c) + f(c) = 0$

20. Ans (D)



$$|iz + 3 - 4i|$$

$$= |i||z - 3i - 4|$$

$$= |z - (4 + 3i)|$$

maximum value of $|iz + 3 - 4i|$
 $= PQ = OP + OQ$
 $= 5 + 4 = 9$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (6)**

$$y^2 = 2 + \cos^2 x$$

$$2yy' = -2\cos x \sin x$$

$$yy' = \frac{-1}{2} \sin 2x$$

$$y^2 + yy'' = -\cos 2x$$

$$yy'' = -(\cos 2x + y^2)$$

$$yy'' = -(\cos 2x + \frac{1}{4y^2} \sin^2 2x)$$

$$y^3 y'' = -(y^2 \cos 2x + \frac{1}{4} \sin^2 2x)$$

$$y^4 + y^3 y'' = -y^2 \cos 2x - \frac{1}{4} \sin^2 2x + y^4$$

$$= (2 + \cos^2 x)(1 - 2\cos^2 x) - \frac{1}{4} \sin^2 2x + (2 + \cos^2 x)^2$$

$$= 2 - 3\cos^2 x - 2\cos^4 x - \frac{1}{4} \sin^2 2x + 4 + \cos^4 x + 4\cos^2 x$$

$$= 6 + \cos^2 x - \cos^4 x - \frac{1}{4} \sin^2 2x$$

$$= 6 + \cos^2 x \sin^2 x - \cos^2 x \sin^2 x = 6$$

2. **Ans (65)**

$$\text{At } x = 0, y = 0$$

$$x + 5y - y^5 = 0$$

$$\Rightarrow 1 + 5y' - 5y^4 y' = 0$$

$$\text{At } x = 0 \text{ \& } y = 0$$

$$\text{Equation of tangent: } y = -\frac{x}{5}, \text{ equation of}$$

$$\text{normal: } y = 5x$$

$$\text{Area} = \frac{1}{2} \times 5 \times 26 = 65$$

3. **Ans (2)**

$$\text{Vector normal to the plane is } \vec{n} = \hat{i} - 3\hat{j} + 2\hat{k}$$

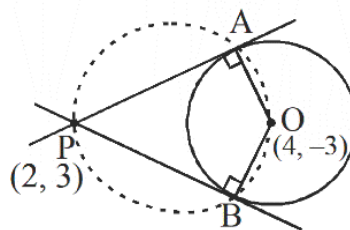
$$\text{and vector along the line is } \vec{V} = 2\hat{i} + \hat{j} - 3\hat{k}$$

$$\text{Now } \sin \theta = \frac{|\vec{x} \cdot \vec{V}|}{|\vec{x}| |\vec{V}|} = \left| \frac{2 - 3 - 6}{\sqrt{14} \sqrt{14}} \right| = \left| \frac{7}{14} \right|$$

$$\text{Hence } \csc \theta = 2 \text{ Ans.}$$

4. **Ans (54)**

Center of the given circle is $O(4, -3)$.



The circumcircle of ΔPAB will circumscribe the quadrilateral $PBOA$ also, hence one of the diameters must be OP .

$\therefore$ Equation of circumcircle of ΔPAB will be

$$(x - 2)(x - 4) + (y - 3)(y + 3) = 0$$

$$\Rightarrow x^2 + y^2 - 6x - 1 = 0 \quad \dots(1)$$

Director circle of given ellipse will be

$$(x + 5)^2 + (y - 3)^2 = 9 + b^2$$

$$\Rightarrow x^2 + y^2 + 10x - 6y + 25 - b^2 = 0 \quad \dots(2)$$

$\therefore$ From (1) and (2), by applying condition of orthogonality, we get

$$2[-3(5) + 0(-3)] = -1 + 25 - b^2$$

$$\Rightarrow -30 = 24 - b^2$$

$$\text{Hence } b^2 = 54 \text{ Ans.}]$$

5.

Ans (2)

$(\pm ae, \frac{b^2}{a})$ are extremities of the latus-rectum having positive ordinates.

$$\Rightarrow a^2 e^2 = -2 \left(\frac{b^2}{a} - 2 \right) \quad \dots(1)$$

$$\text{But } b^2 = a^2 (1 - e^2) \quad \dots(2)$$

$\therefore$ From (1) and (2), we get $a^2 e^2 - 2ae^2 + 2a - 4 = 0$

$$\Rightarrow ae^2 (a - 2) + 2(a - 2) = 0$$

$$\therefore (ae^2 + 2)(a - 2) = 0$$

$$\text{Hence } a = 2$$

6. **Ans (12)**

$$9y^2 = x^3 \Rightarrow 9b^2 = a^3$$

$$y' = \frac{3x^2}{18y} = \frac{x^2}{6y} \Big|_{(a,b)} = \frac{a^2}{6b}$$

$$\text{Slope of normal} = \frac{-6b}{a^2}$$

$$\therefore \left| \frac{6b}{a^2} \right| = 1$$

$$|6b| = a^4$$

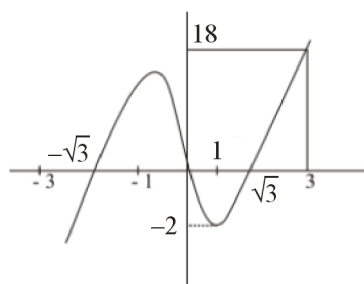
$$b^2 = \frac{a^4}{36} \text{ \& } b^2 = \frac{a^3}{9}$$

$$\therefore \frac{a^4}{36} = \frac{a^3}{9} \Rightarrow a = 4$$

$$\therefore b = \sqrt{\frac{4^3}{9}} = \frac{8}{3}$$

$$\therefore a + 3b = 12$$

7. **Ans (18)**



$$x^4 - 10x^2 + 9 \leq 0$$

$$(x^2 - 9)(x^2 - 1) \leq 0$$

$$\text{hence } -3 < x < -1 \text{ or } 1 < x < 3$$

$$\text{now } f(x) = x^3 - 3x$$

$$f'(x) = 3x^2 - 3 = 0$$

$$x = \pm 1$$

$$\text{maximum occurs when } x = 3$$

$$f(3) = 18$$

8. **Ans (893)**

From triangle inequality

$$\log_{10} 12 + \log_{10} 75 > \log_{10} n$$

$$\log_{10} 900 > \log_{10} n \Rightarrow n < 900 \quad \dots(1)$$

$$\text{also } \log_{10} 12 + \log_{10} n > \log_{10} 75$$

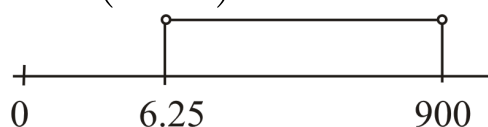
$$\log_{10} 12n > \log_{10} 75$$

$$12n > 75 \quad \dots(2)$$

$$n > \frac{75}{12} \text{ or } n > \frac{25}{4}$$

from (1) & (2)

$$\Rightarrow n \in \left(\frac{25}{4}, 900 \right)$$



$$\text{Hence no. of values} = 900 - 7 = 893$$

9. **Ans (6)**

$$P(x) = ax^5 + bx^4 + cx^3$$

$$P'(x) = 5ax^4 + 4bx^3 + 3cx^2$$

$$\left. \begin{aligned} P'(1) &= 5a + 4b + 3c = 0 \\ P'(-1) &= 5a - 4b + 3c = 0 \end{aligned} \right\} b = 0$$

$$\lim_{x \rightarrow 0} \left(\frac{P(x)}{x^3} - 2 \right) = 4$$

$$\therefore c - 2 = 4 \Rightarrow c = 6$$

$$b = 0 \quad c = 6 \text{ then } a = \frac{-18}{5}$$

$$\therefore P(x) = \frac{-18}{5}x^5 + 6x^3$$

$$P'(x) = -18(x^4 - x^2)$$

$$A = [2, 3]$$

$P'(x)$ is decreasing in A

$$\therefore m = P(3) = -18(81 - 9) = -18 \times 72$$

$$M = P(2) = -18(16 - 4) = -18 \times 12$$

$$\therefore \frac{m}{M} = 6$$

10. **Ans (432)**

Number of numbers having '1' repeated twice

$$= {}^3C_1 \times 9 \times 8 = 216$$

Number of numbers having one digit repeated other than '1'

$$= {}^9C_1 \times {}^3C_2 \times 1 \times 8$$

$$= 9 \times 3 \times 8 = 216$$

$$\therefore \text{Total number of numbers} = 216 + 216 = 432$$